



CIRCUITOS DIGITALES Y MICROCONTROLADORES 2026

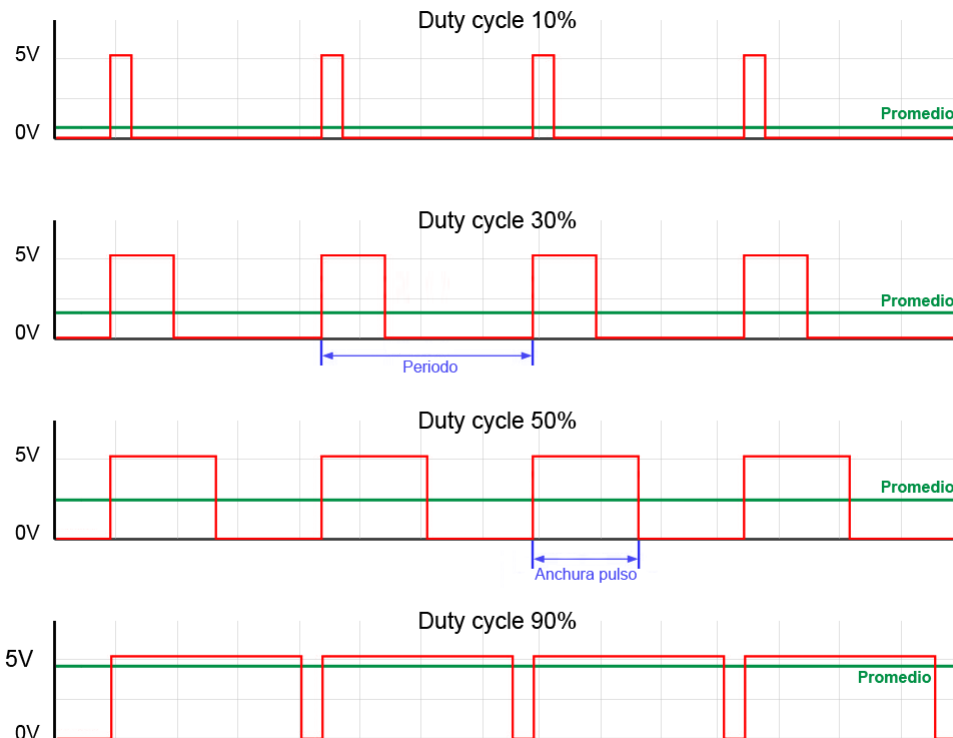
Facultad de Ingeniería
UNLP

PERIFERICOS TIMER
Modos PWM

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Señales PWM (Pulse Width Modulation)

- Es una técnica de modulación digital donde la información útil de la señal se encuentra en el ancho del pulso.
- Esto permite que a partir de una señal digital de ancho de pulso variable se pueda obtener cierta información en forma analógica.



- Es decir, permite controlar dispositivos analógicos por medio de salidas digitales o generar señales analógicas a partir de la conversión digital-analógica (D/A)
- Ejemplos: control de velocidad de motores DC, control de intensidad de lámparas, control de color en LED RGB.

PWM – Timer 0

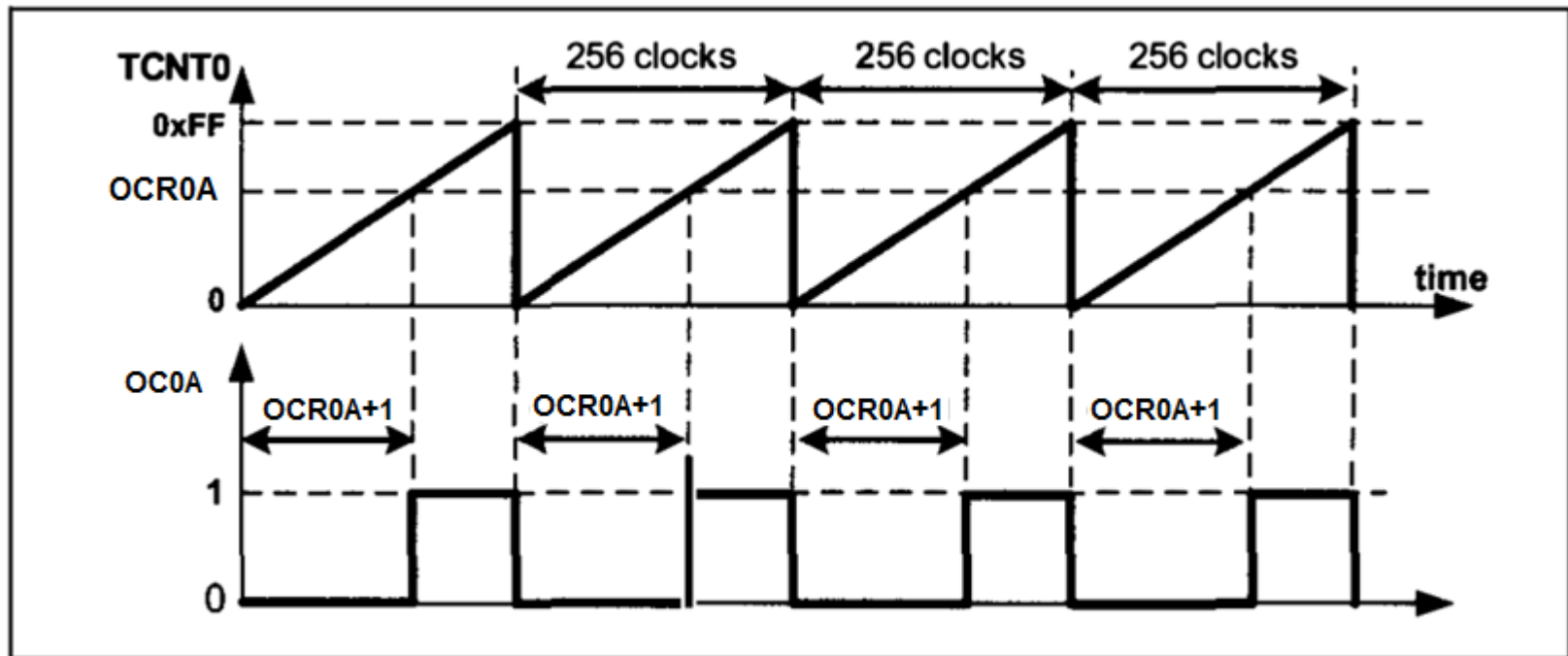
Table 14-8. Waveform Generation Mode Bit Description

Mode	WGM02	WGM01	WGM00	Timer/Counter Mode of Operation	TOP	Update of OCRx at	TOV Flag Set on ⁽¹⁾⁽²⁾
0	0	0	0	Normal	0xFF	Immediate	MAX
1	0	0	1	PWM, Phase Correct	0xFF	TOP	BOTTOM
2	0	1	0	CTC	OCRA	Immediate	MAX
3	0	1	1	Fast PWM	0xFF	BOTTOM	MAX
4	1	0	0	Reserved	–	–	–
5	1	0	1	PWM, Phase Correct	OCRA	TOP	BOTTOM
6	1	1	0	Reserved	–	–	–
7	1	1	1	Fast PWM	OCRA	BOTTOM	TOP

Notes: 1. MAX = 0xFF
2. BOTTOM = 0x00

Generador PWM – Timer 0

- Modo Fast PWM modo 3: operación



Modo Fast PWM Modo Invertido

PWM – Timer 0

Table 14-3. Compare Output Mode, Fast PWM Mode⁽¹⁾

COM0A1	COM0A0	Description
0	0	Normal port operation, OC0A disconnected.
0	1	WGM02 = 0: Normal Port Operation, OC0A Disconnected. WGM02 = 1: Toggle OC0A on Compare Match.
1	0	Clear OC0A on Compare Match, set OC0A at BOTTOM, (non-inverting mode).
1	1	Set OC0A on Compare Match, clear OC0A at BOTTOM, (inverting mode).

Modo Fast PWM: invertido, no invertido

PWM – Timer 0

- Utilizaremos el generador de onda y la salida de comparación con las siguientes posibilidades:

COM0A1=1
COM0A0=0
no invertido

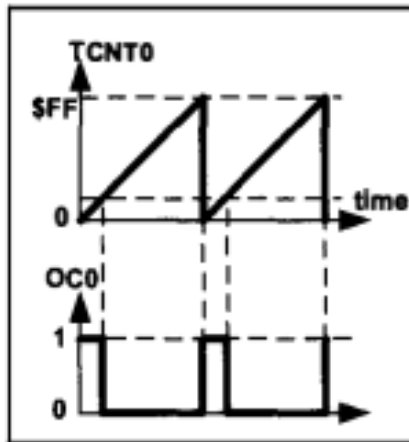


Figure 16-13A. Non-inverted

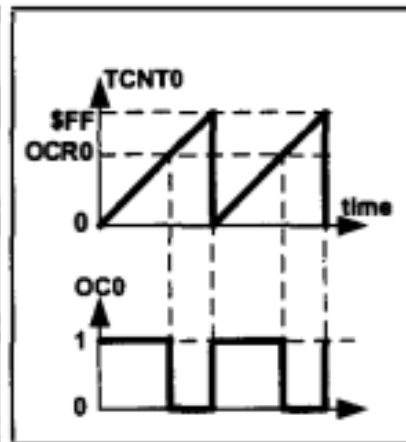


Figure 16-13B. Non-inverted

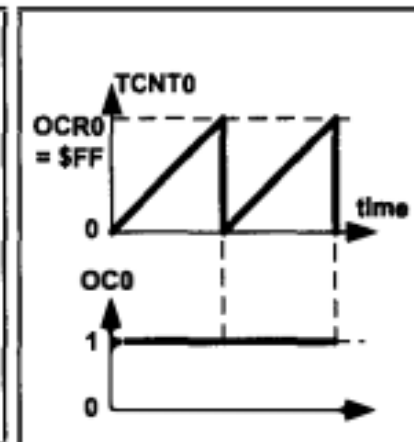


Figure 16-13C. Non-inverted

COM0A1=1
COM0A0=1
invertido

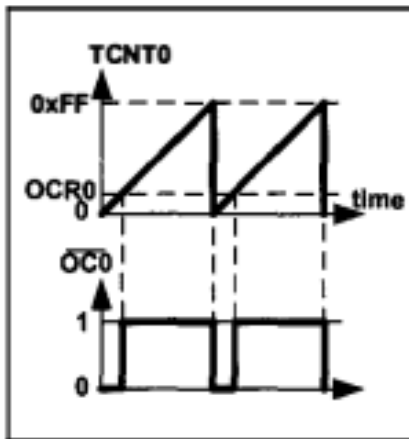


Figure 16-14A. Inverted

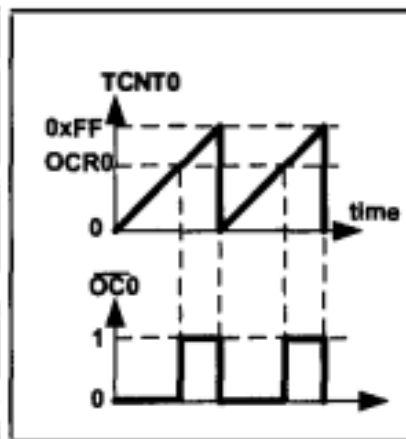


Figure 16-14B. Inverted

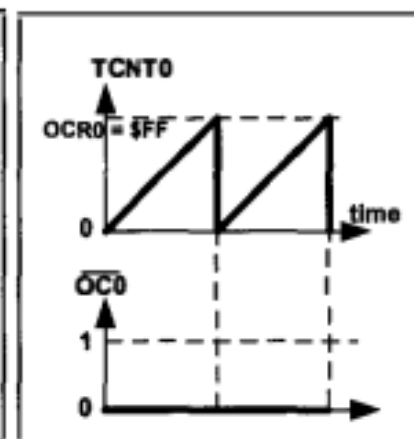
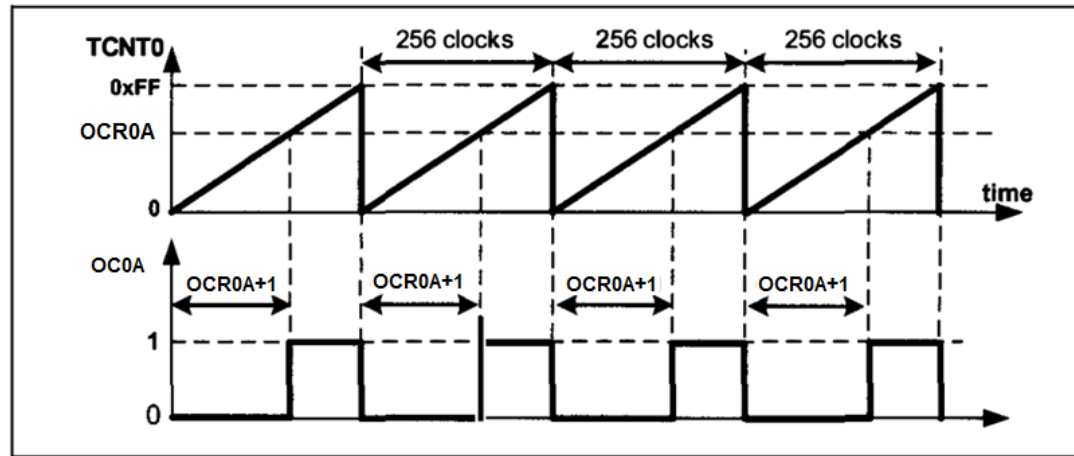


Figure 16-14C. Inverted

PWM – Timer 0



- Frecuencia de la señal generada Modo 3:

$$\left. \begin{aligned} F_{\text{generated wave}} &= \frac{F_{\text{timer clock}}}{256} \\ F_{\text{timer clock}} &= \frac{F_{\text{oscillator}}}{N} \end{aligned} \right\} \Rightarrow F_{\text{generated wave}} = \frac{F_{\text{oscillator}}}{256 \times N}$$

- Ciclo de trabajo: $p = \delta/T$

$$\text{Duty Cycle} = \frac{\text{OCR0} + 1}{256} \times 100$$

Modo no invertido

$$\text{Duty Cycle} = \frac{255 - \text{OCR0}}{256} \times 100$$

Modo invertido

PWM – Timer 0

- Ejemplo: Generar pwm con $f=7812.5\text{Hz}$ y $CT=37.5\%$
 - Supongamos $f_{clk}=16\text{MHz}$

⇒ $\text{duty cycle} = (\text{OCR0A} + 1) * 100 / 256 = 37.5\%$ despejando $\text{OCR0A} = 95$

⇒ si $N=8$ $f_{pwm} = f_{clk} / (256 * N) = 7812.5\text{Hz}$

⇒ Código:

```
DDRD  |= (1<<6);    //PIN PD6 corresponde a la salida OC0A
OCR0A=95;
TCCR0A=(1<<COM0A1) | (1<<WGM01) | (1<<WGM00);    //modo Fast-non inv.
TCCR0B=(1<<CS01);    //prediv 8
```


PWM – Timer 1 (16 bits)

- Modos Fast PWM

Mode	WGM13	WGM12	WGM11	WGM10	Timer/Counter Mode of Operation	Top	Update of OCR1x	TOV1 Flag Set on
0	0	0	0	0	Normal	0xFFFF	Immediate	MAX
1	0	0	0	1	PWM, Phase Correct, 8-bit	0x00FF	TOP	BOTTOM
2	0	0	1	0	PWM, Phase Correct, 9-bit	0x01FF	TOP	BOTTOM
3	0	0	1	1	PWM, Phase Correct, 10-bit	0x03FF	TOP	BOTTOM
4	0	1	0	0	CTC	OCR1A	Immediate	MAX
5	0	1	0	1	Fast PWM, 8-bit	0x00FF	TOP	TOP
6	0	1	1	0	Fast PWM, 9-bit	0x01FF	TOP	TOP
7	0	1	1	1	Fast PWM, 10-bit	0x03FF	TOP	TOP
8	1	0	0	0	PWM, Phase and Frequency Correct	ICR1	BOTTOM	BOTTOM
9	1	0	0	1	PWM, Phase and Frequency Correct	OCR1A	BOTTOM	BOTTOM
10	1	0	1	0	PWM, Phase Correct	ICR1	TOP	BOTTOM
11	1	0	1	1	PWM, Phase Correct	OCR1A	TOP	BOTTOM
12	1	1	0	0	CTC	ICR1	Immediate	MAX
13	1	1	0	1	Reserved	–	–	–
14	1	1	1	0	Fast PWM	ICR1	TOP	TOP
15	1	1	1	1	Fast PWM	OCR1A	TOP	TOP

PWM – Timer 1 (16 bits)

- 5 Modos Fast PWM dependiendo del valor TOP del contador:

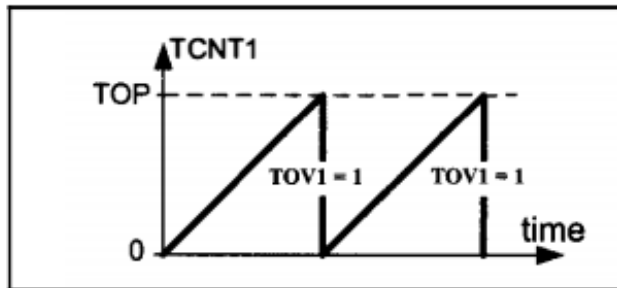


Figure 16-24. Fast PWM Mode

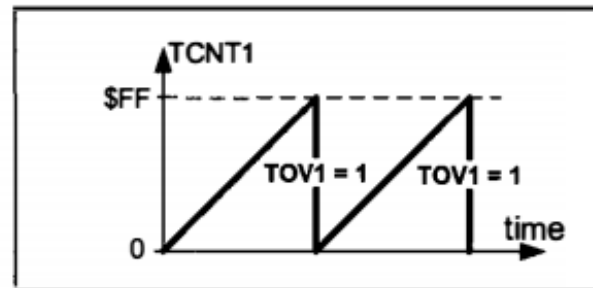


Figure 16-25. TOV in Mode 5

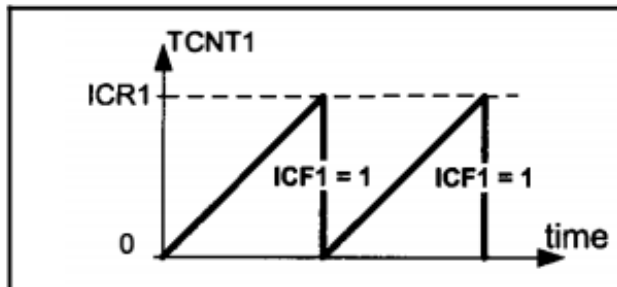


Figure 16-26. Mode 14

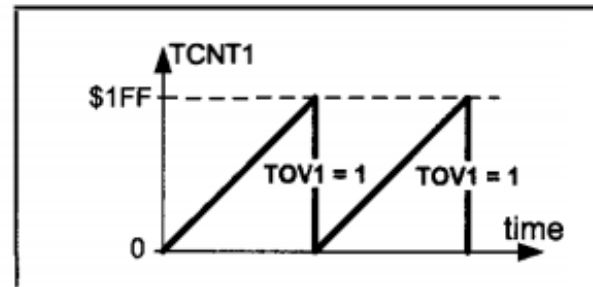


Figure 16-27. TOV in Mode 6

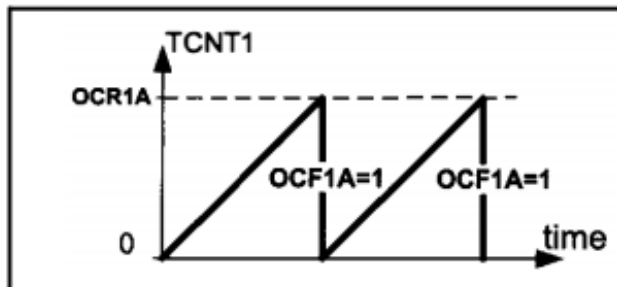


Figure 16-28. Mode 15

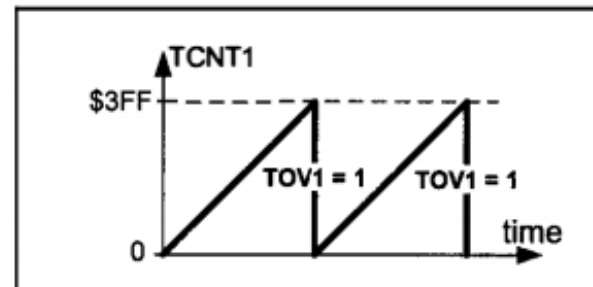


Figure 16-29. TOV in Mode 7

PWM – Timer 1 (16 bits)

- Modo no invertido

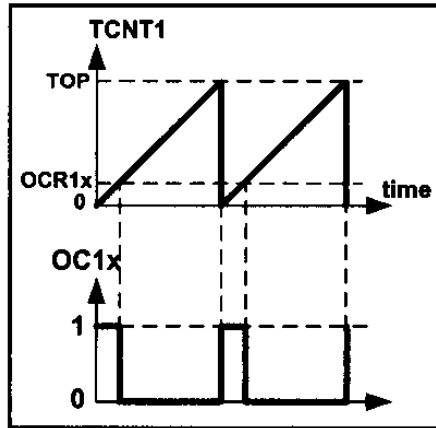


Figure 16-32A. Non-inverted

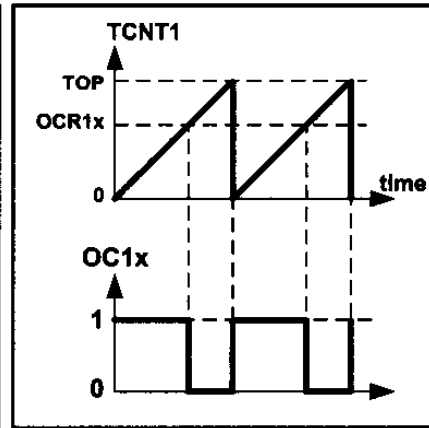


Figure 16-32B. Non-inverted

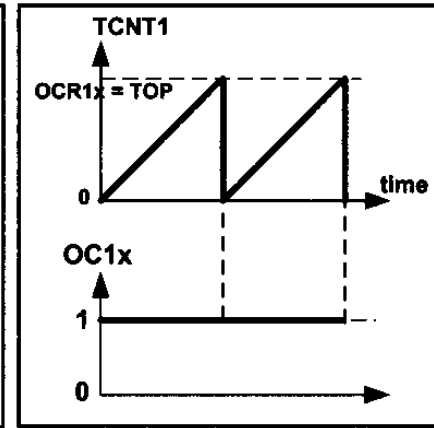


Figure 16-32C. Non-inverted

- Ídem Modo invertido
- Frecuencia fast pwm:

Número de bits de un PWM

$$R_{FPWM} = \frac{\log(TOP + 1)}{\log(2)}$$

$$\left. \begin{aligned} F_{\text{generated wave}} &= \frac{F_{\text{timer clock}}}{Top + 1} \\ F_{\text{timer clock}} &= \frac{F_{\text{oscillator}}}{N} \end{aligned} \right\} \Rightarrow F_{\text{generated wave}} = \frac{F_{\text{oscillator}}}{(Top + 1) \times N}$$

PWM – Timer 1 (16 bits)

- Ciclo de trabajo: $p = \delta/T$

No invertido:

$$\text{Duty Cycle} = \frac{\text{OCR1x} + 1}{\text{Top} + 1} \times 100$$

Invertido:

$$\text{Duty Cycle} = \frac{\text{Top} - \text{OCR1x}}{\text{Top} + 1} \times 100$$

Ejemplo: para ciclo de trabajo 75%, el OCR1A debe ser:

- 191 (en modo 5, no inv)
- 383 (en modo 6, no inv)
- 767 (en modo 7, no inv)

PWM – Timer 1 (16 bits)

- Ejemplo: Generar PWM no inv. M5, $f=31.250\text{Hz}$ y $\text{cicloT}=75\%$.
 - Sup $f_{\text{clk}}=8\text{MHz}$

```
DDRB |= (1<<PB1);    //PIN PB1 corresponde a la salida OC1A
OCR1A=191;
TCCR1A=(1<<COM1A1) | (1<<WGM10); //modo 5 Fast-non inv.
TCCR1B=(1<<CS10) | (1<<WGM12);    //prediv 1
```

- Notar que las posibles frecuencias a generar dependen del prescalador

Prescaler	1	1:8	1:64	1:256	1:1024
Mode = 5	$\frac{F_{\text{oscillator}}}{1 \times 256}$	$\frac{F_{\text{oscillator}}}{8 \times 256}$	$\frac{F_{\text{oscillator}}}{64 \times 256}$	$\frac{F_{\text{oscillator}}}{256 \times 256}$	$\frac{F_{\text{oscillator}}}{1024 \times 256}$
Mode = 6	$\frac{F_{\text{oscillator}}}{1 \times 512}$	$\frac{F_{\text{oscillator}}}{8 \times 512}$	$\frac{F_{\text{oscillator}}}{64 \times 512}$	$\frac{F_{\text{oscillator}}}{256 \times 512}$	$\frac{F_{\text{oscillator}}}{1024 \times 512}$
Mode = 7	$\frac{F_{\text{oscillator}}}{1 \times 1024}$	$\frac{F_{\text{oscillator}}}{8 \times 1024}$	$\frac{F_{\text{oscillator}}}{64 \times 1024}$	$\frac{F_{\text{oscillator}}}{256 \times 1024}$	$\frac{F_{\text{oscillator}}}{1024 \times 1024}$

PWM – Timer 1 (16 bits)

- Para generar otras frecuencias se utilizan los modos 14 o 15 donde TOP se especifica con los registros ICR1 o OCR1A respectivamente.
- Por ejemplo: generar fast pwm, modo 14, $f=80\text{kHz}$

$$80\text{kHz}=8\text{MHz} / N*(\text{TOP}+1) \Rightarrow / N*(\text{TOP}+1) = 8\text{MHz}/80\text{kHz} = 100$$

$$\text{Si } N=1 \text{ (CS1[2:0]=001)}$$

$$\text{TOP}+1=100$$

$$\text{TOP}=99$$

- Para ciclo de trabajo 20%

$$20=(\text{OCR1x}+1)*100 / (99+1) \Rightarrow \text{OCR1x}+1=20 \Rightarrow \text{OCR1x}=19$$

```
DDRB |= (1<<PB1);    //PIN PB1 corresponde a la salida OC1A
ICR1A=99;
OCR1A=19;
TCCR1A=(1<<COM1A1) | (1<<WGM11); //modo 14 Fast-non inv.
TCCR1B=(1<<CS10) | (1<<WGM13) | (1<<WGM12); //prediv 1
```

PWM Fase correcta– Timer 0

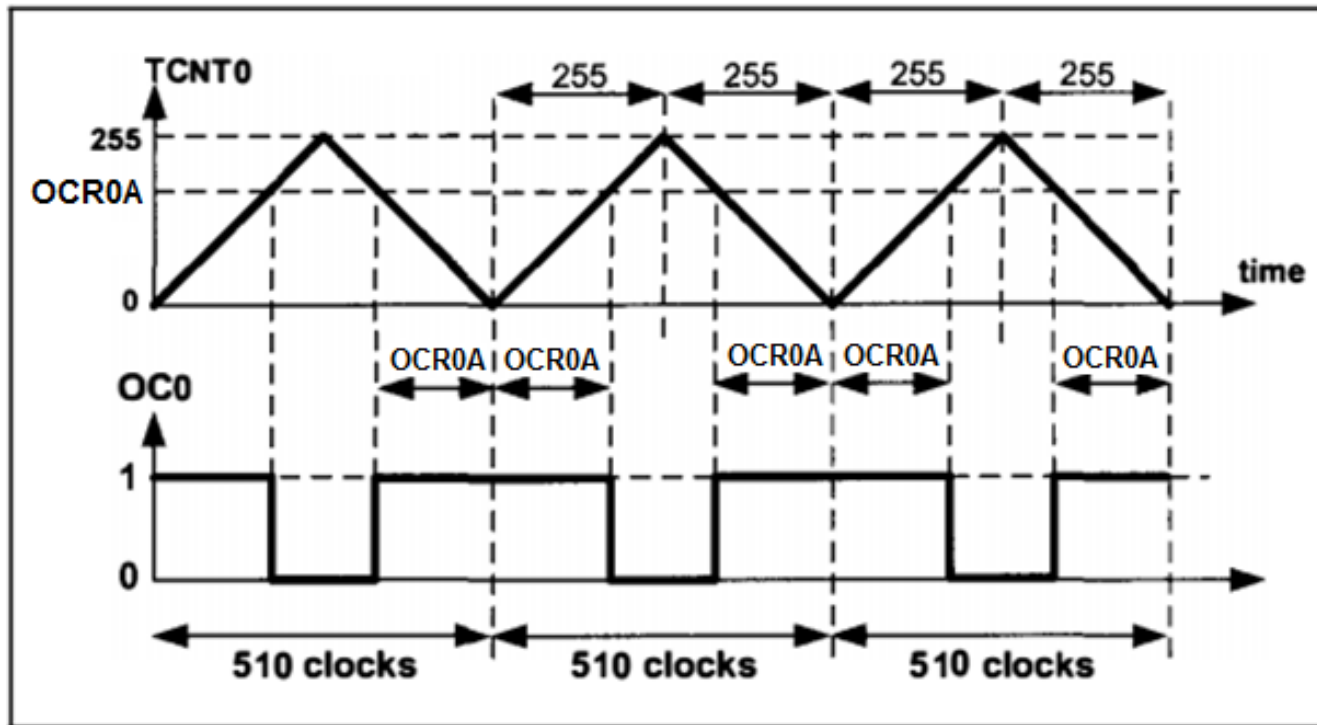
Table 14-8. Waveform Generation Mode Bit Description

Mode	WGM02	WGM01	WGM00	Timer/Counter Mode of Operation	TOP	Update of OCRx at	TOV Flag Set on ⁽¹⁾⁽²⁾
0	0	0	0	Normal	0xFF	Immediate	MAX
1	0	0	1	PWM, Phase Correct	0xFF	TOP	BOTTOM
2	0	1	0	CTC	OCRA	Immediate	MAX
3	0	1	1	Fast PWM	0xFF	BOTTOM	MAX
4	1	0	0	Reserved	–	–	–
5	1	0	1	PWM, Phase Correct	OCRA	TOP	BOTTOM
6	1	1	0	Reserved	–	–	–
7	1	1	1	Fast PWM	OCRA	BOTTOM	TOP

- Notes: 1. MAX = 0xFF
2. BOTTOM = 0x00

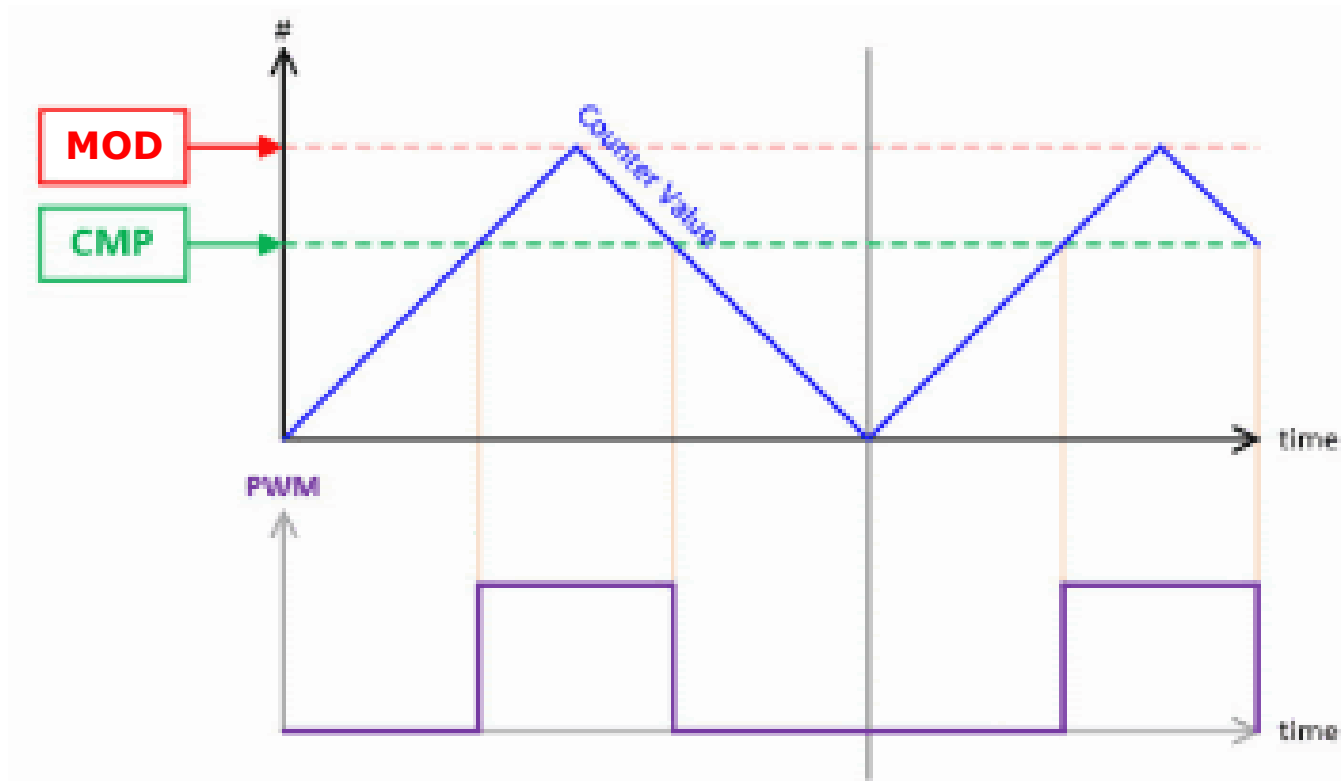
PWM Fase correcta– Timer 0

- Modo Fase correcta



$$\left. \begin{aligned} F_{\text{generated wave}} &= \frac{F_{\text{timer clock}}}{510} \\ F_{\text{timer clock}} &= \frac{F_{\text{oscillator}}}{N} \end{aligned} \right\} \Rightarrow F_{\text{generated wave}} = \frac{F_{\text{oscillator}}}{510 \times N}$$

PWM Fase correcta– Timer 0



PWM Fase correcta– Timer 1 (16 bits)

- Modos PWM

Mode	WGM13	WGM12	WGM11	WGM10	Timer/Counter Mode of Operation	Top	Update of OCR1x	TOV1 Flag Set on
0	0	0	0	0	Normal	0xFFFF	Immediate	MAX
1	0	0	0	1	PWM, Phase Correct, 8-bit	0x00FF	TOP	BOTTOM
2	0	0	1	0	PWM, Phase Correct, 9-bit	0x01FF	TOP	BOTTOM
3	0	0	1	1	PWM, Phase Correct, 10-bit	0x03FF	TOP	BOTTOM
4	0	1	0	0	CTC	OCR1A	Immediate	MAX
5	0	1	0	1	Fast PWM, 8-bit	0x00FF	TOP	TOP
6	0	1	1	0	Fast PWM, 9-bit	0x01FF	TOP	TOP
7	0	1	1	1	Fast PWM, 10-bit	0x03FF	TOP	TOP
8	1	0	0	0	PWM, Phase and Frequency Correct	ICR1	BOTTOM	BOTTOM
9	1	0	0	1	PWM, Phase and Frequency Correct	OCR1A	BOTTOM	BOTTOM
10	1	0	1	0	PWM, Phase Correct	ICR1	TOP	BOTTOM
11	1	0	1	1	PWM, Phase Correct	OCR1A	TOP	BOTTOM
12	1	1	0	0	CTC	ICR1	Immediate	MAX
13	1	1	0	1	Reserved	–	–	–
14	1	1	1	0	Fast PWM	ICR1	TOP	TOP
15	1	1	1	1	Fast PWM	OCR1A	TOP	TOP

PWM Fase correcta – Timer 1 (16 bits)

- Modo Fase correcta

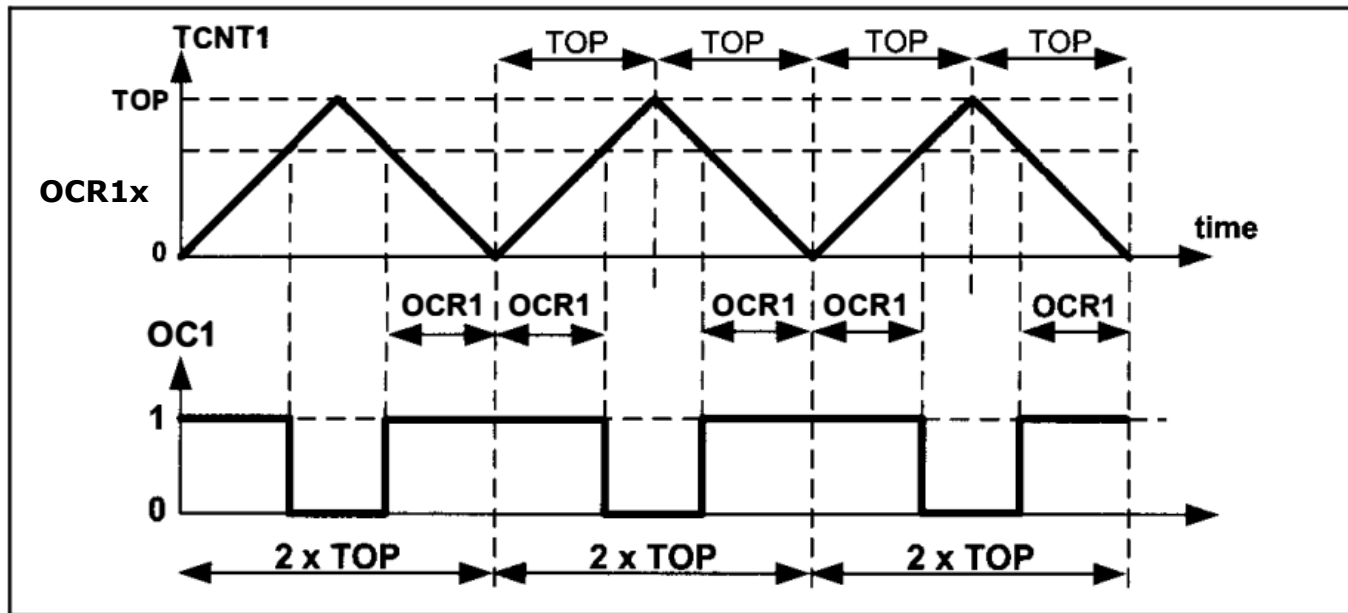


Figure 16-46. Timer/Counter 1 Phase Correct PWM Mode

$$\begin{aligned}
 F_{\text{generated wave}} &= \frac{F_{\text{timer clock}}}{2 \times \text{Top}} \\
 F_{\text{timer clock}} &= \frac{F_{\text{oscillator}}}{N}
 \end{aligned}
 \left. \vphantom{\begin{aligned} F_{\text{generated wave}} &= \frac{F_{\text{timer clock}}}{2 \times \text{Top}} \\ F_{\text{timer clock}} &= \frac{F_{\text{oscillator}}}{N} \end{aligned}} \right\} \Rightarrow F_{\text{generated wave}} = \frac{F_{\text{oscillator}}}{2 \times N \times \text{Top}}$$

PWM Fase correcta – Timer 1 (16 bits)

- Modo Fase correcta

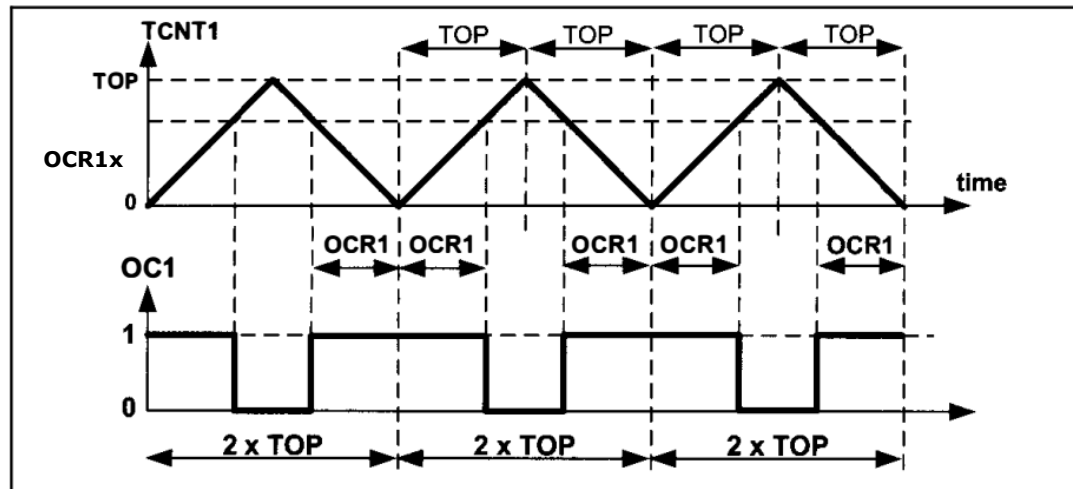


Figure 16-46. Timer/Counter 1 Phase Correct PWM Mode

- No Invertido:

$$\text{Duty Cycle} = \frac{2 \times \text{OCR1A}}{2 \times \text{Top}} \times 100 \quad \Rightarrow \quad \text{Duty Cycle} = \frac{\text{OCR1A}}{\text{Top}} \times 100$$

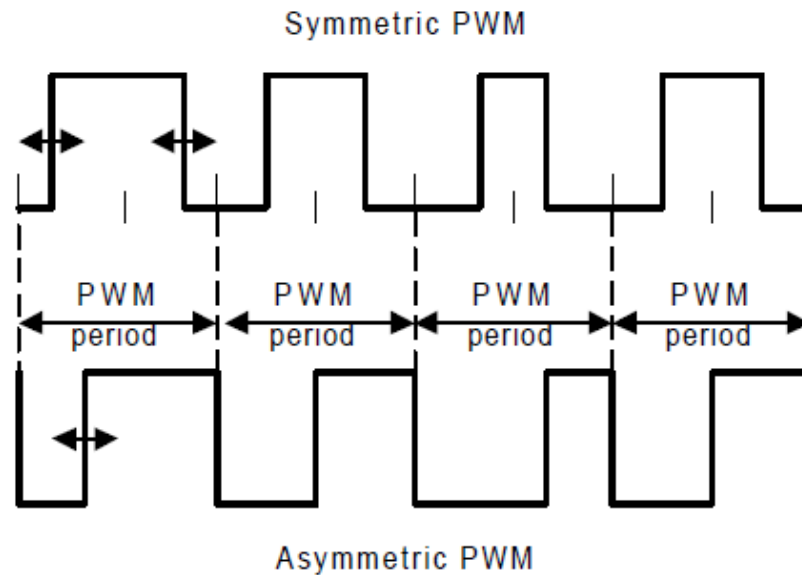
- Invertido:

$$\text{Duty Cycle} = \frac{2 \times \text{Top} - 2 \times \text{OCR1A}}{2 \times \text{Top}} \times 100 \quad \Rightarrow \quad \text{Duty Cycle} = \frac{\text{Top} - \text{OCR1A}}{\text{Top}} \times 100$$

Notar que puede obtenerse el rango 0 a 100% en un mismo modo

PWM Fase correcta – Timer 1 (16 bits)

- Diferencias entre modos:



- Modo simétrico se utiliza preferentemente para el control de motores de múltiples fases como los motores de inducción o los BLDC (generalmente 3 o 6 fases).
- En fase correcta, el centro del pulso no varía, con lo cual puede utilizarse como instante para sensar la corriente.
- En motores de más de una fase el contenido armónico generado es menor.

PWM Fase correcta– Timer 1 (16 bits)

- En modo Fase correcta, la zona muerta de las señales PWM diferenciales puede controlarse fácilmente desplazando los valores de los registros:

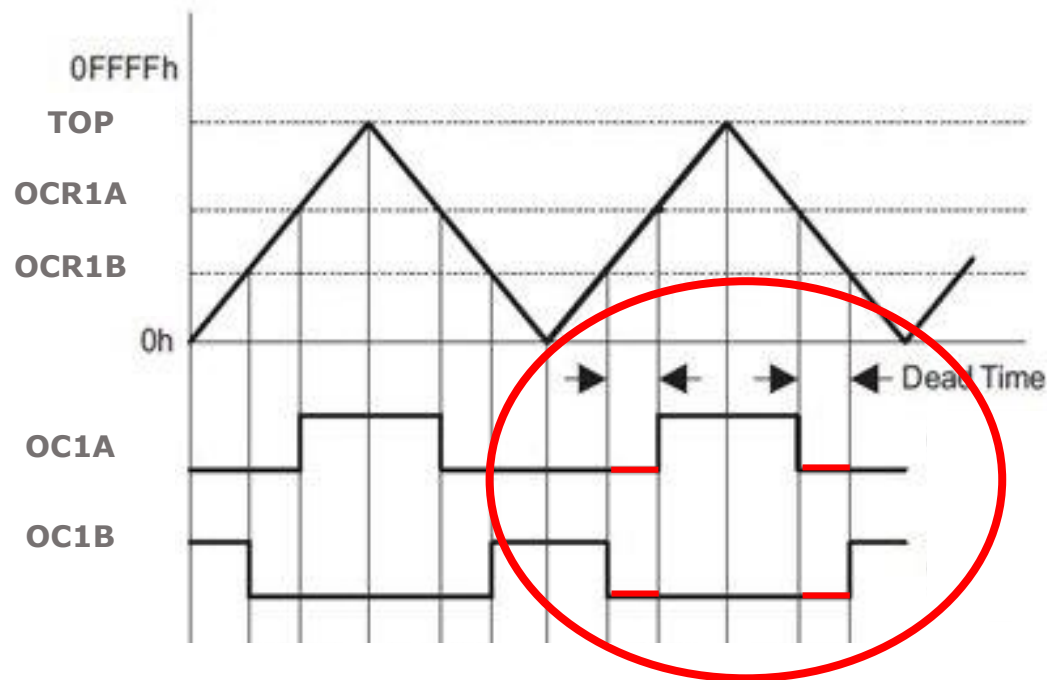


Figure 12-9. Output Unit in Up/Down Mode